

BROCKVILLE, ON, Canada

WMO: 711610

Lat: 44.6354N

Lon: 75.7516W

Elev: 395

StdP: 14.49

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-5.9	-19.4	1.9	-11.0	-13.4	2.7	-4.7	25.0	25.4	21.9	26.6	5.5	310	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	85.4	70.8	82.8	69.3	80.5	68.4	73.8	82.1	72.1	79.6	70.6	77.3	9.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	116.4	79.1	69.5	110.2	76.6	68.1	104.7	74.9	37.7	82.3	36.1	79.8	34.7	77.6	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.7	15.5	-22.8	90.2	5.4	3.1	-26.7	92.5	-29.9	94.3	-32.9	96.1	-36.8	98.3		
(5)			-23.1	77.3	5.3	1.9	-26.9	78.6	-30.0	79.7	-33.0	80.8	-36.9	82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	17.2	20.1	29.8	43.5	57.0	63.8	69.5	67.6	60.3	48.5	36.2	24.7		
	DBStd	20.34	12.74	11.70	11.24	8.39	8.00	5.97	5.21	4.99	7.59	8.34	9.56	11.88		
	HDD50	4083	1017	836	632	229	26	0	0	0	6	130	422	785		
	HDD65	7691	1482	1256	1091	647	268	92	17	31	182	512	864	1249		
	CDD50	2254	0	0	6	34	244	414	603	546	315	85	7	0		
	CDD65	385	0	0	0	1	21	55	155	112	40	1	0	0		
	CDH74	3005	0	0	2	26	226	420	1267	741	315	8	0	0		
	CDH80	622	0	0	0	3	41	64	331	131	52	0	0	0		
	WSAvg	7.1	7.8	8.1	8.1	8.5	7.2	6.4	6.1	5.8	6.0	7.2	7.4	7.2		
	PrecAvg	38.7	3.0	2.4	3.0	3.2	3.0	2.9	3.1	3.6	3.6	3.2	4.0	3.6		
(15) Precipitation	PrecMax	48.0	5.9	5.7	4.9	5.9	4.8	5.4	7.8	7.1	7.4	5.4	6.5	5.9		
	PrecMin	29.8	1.0	0.6	1.1	1.1	1.2	0.6	1.3	0.9	1.1	1.2	2.0	1.5		
	PrecStd	4.0	1.3	1.3	1.1	1.3	1.2	1.2	1.5	1.4	1.5	1.3	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	51.3	69.4	78.4	84.4	85.4	90.4	87.2	84.8	75.1	65.1	55.2		
		MCWB	48.5	48.2	59.2	63.0	68.9	71.0	73.7	70.2	71.9	65.5	55.4	50.4		
	2%	DB	43.7	44.4	55.8	68.2	79.9	81.6	86.8	83.3	80.9	70.4	58.9	48.1		
		MCWB	40.7	41.0	47.6	55.4	65.5	67.9	71.3	70.1	70.1	62.3	51.5	45.9		
	5%	DB	38.3	40.2	49.7	62.3	75.9	78.8	83.6	80.8	77.7	66.0	54.5	44.3		
		MCWB	35.9	36.7	43.1	51.9	62.9	65.6	69.8	69.0	67.9	58.1	48.9	42.4		
	10%	DB	34.6	36.3	44.6	57.7	71.8	75.9	81.0	78.4	74.2	61.3	50.7	39.4		
		MCWB	32.6	33.7	38.8	48.3	60.4	64.3	68.7	67.8	66.0	55.6	46.1	36.9		
	0.4%	WB	48.9	48.1	59.2	63.3	71.5	73.9	77.2	74.7	74.3	67.7	56.5	52.2		
		MCDB	50.6	50.4	69.1	74.8	81.9	82.6	87.2	83.0	82.2	72.3	61.9	53.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.6	41.4	48.7	57.7	67.9	71.0	74.5	72.9	72.0	63.5	53.1	46.1		
		MCDB	42.8	44.5	54.5	65.1	77.0	79.0	83.0	80.3	79.1	68.7	58.1	47.9		
	5%	WB	36.0	36.8	43.9	53.1	65.2	68.6	72.5	71.1	69.5	59.6	49.7	42.0		
		MCDB	38.0	39.3	48.8	60.1	73.1	74.9	80.0	77.9	75.3	64.6	53.8	44.5		
	10%	WB	32.9	33.8	39.6	49.6	62.4	66.5	70.7	69.4	67.2	56.5	46.2	37.3		
		MCDB	34.5	36.1	43.6	56.9	68.6	72.6	77.5	75.3	72.5	60.4	50.7	39.3		
	5% DB	MDBR	17.0	16.7	16.6	18.4	20.1	18.6	19.6	19.0	20.1	16.3	15.1	13.7		
		MCDBR	17.6	18.4	22.1	27.2	25.7	23.0	23.1	21.8	22.8	20.8	21.4	16.6		
		MCWBR	16.7	16.3	15.2	16.5	14.0	12.1	11.0	11.0	12.9	13.4	16.0	15.6		
		MCWBR	17.4	18.5	20.1	23.6	21.5	17.8	19.1	17.7	20.3	19.0	19.1	17.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.7	17.0	15.1	16.8	13.7	10.7	10.4	10.2	12.7	13.7	15.5	16.2		
	Clear-Sky Solar Irradiance	taub	0.285	0.292	0.320	0.349	0.379	0.413	0.423	0.412	0.365	0.344	0.306	0.281		
		taud	2.379	2.362	2.422	2.405	2.363	2.280	2.256	2.292	2.411	2.466	2.511	2.449		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	265	285	291	290	283	272	267	266	272	263	257	256		
		Edh at Noon	24	29	32	35	38	41	42	39	32	26	21	20		
(44) All-Sky Solar Radiation	RadStd		430	717	1136	1401	1669	1791	1832	1613	1280	759	475	328		
			43	72	93	167	172	142	157	74	108	71	52	38		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page